Integrated Path Planning and Tracking Control of an AUV: A Unified Receding Horizon Optimization Approach

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***Abstract*—This paper attempts to develop a unified receding horizon optimization (RHO) scheme for the integrated path planning and tracking control of an autonomous underwater vehicle (AUV). Considering that the effective sensing range of onboard sensors is practically short, we formulate the path planning into RHO problems with the spline path template. The planned path is subsequently viewed as the state trajectory of a virtual reference system having the same kinematic and dynamic properties as the AUV's. Appropriately constructed error dynamics makes the AUV tracking control equivalent to the regulation problem of the error dynamic system, which facilitates the derivation of theoretical results via nonlinear MPC techniques. The model predictive control (MPC) tracking controller is designed so that closed-loop stability can be ensured. Due to the inherent RHO nature, both the path planning and tracking control are incorporated into an unified scheme. Simulation studies are conducted using a realistic dynamic model of the *Falcon* AUV, which was created in our previous experimental work. The simulation results demonstrate the effectiveness of the proposed control algorithm.**

***Index Terms*—Autonomous underwater vehicle (AUV), dynamic path planning, nonlinear model predictive control (MPC), receding horizon optimization (RHO), trajectory tracking.**

I. Introduction

OVER the past few decades, autonomous underwater vehicles (AUVs) have been receiving significant amount of attention from academia and from the industry due to their great potential to reduce risks in deep oceanic exploration and exploitation of underwater resources [1]. Recently, offshore petrochemical industry companies moving their drilling platforms to the seabed initiated the enthusiasm of designing and manufacturing operational AUVs for inspection, maintenance, and repair purposes [2]–[4]. All these ambitions make AUV research an active field in robotics and the science of system control. The salient feature of AUVs is the ability to perform appropriate actions to achieve predefined objectives without interventions of human pilots [5], [6]. Such a high level of autonomy, in turn, justifies an elaborately designed control system for the AUV. The highly nonlinear, time-varying hydrodynamic properties of the AUV [7], however, challenge the designs of a motion controller.

The majority of conventional marine vessel controller designs take the classical linearization regime followed by the extensively studied linear control techniques, e.g., proportional-integral-derivative (PID) [7]. However, AUVs are small multipurpose underwater vehicles, operating over numerous working points, which complicates the application of linear control theory. Nonlinear effects from hydrodynamic damping and lift, added mass, Coriolis, and centripetal forces may result in significantly degraded performance even though AUVs are operating slightly away from the preset working point. Therefore, applications of advanced control techniques have been constantly catching the researchers' eyes [8], [9]. As listed in [10], a wide range of advanced control theories have been successfully applied to the AUV motion control.

AUV motion control problems are classified into four categories: vertical and horizontal plane control, attitude control, trajectory tracking, and path following. Essentially, the first two fall into the classic control problem of stabilizing the system state to the equilibrium point, while the third and fourth appear more complicated due to the time-varying nature of desired output. For the tracking of piecewise linear paths, the line-of-sight (LOS) scheme is often used [11]. To stabilize the cross-track error in the LOS scheme, conventional PID [7], LQG [12], and nonlinear PID control techniques [13] have been applied to AUVs. For tracking of time-parameterized curves, the Lyapunov-based backstepping technique can be applied cf., [14]. Also, due to the insensitivity to parametric uncertainty, the sliding mode control [15] is suitable for the AUV tracking control as well. Nevertheless, the above-mentioned control methods lack the capability of handling system constraints that are ubiquitous, typically in terms of actuator limits. This motivates researchers to investigate the model predictive control (MPC) for AUV tracking problems.

The beauty of MPC lies in the fact that it can conveniently handle nonlinear multiple-input multiple-output system control problems, and explicitly take system constraints into

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consideration [16]–[19]. Regarding AUV applications, the linear MPC formulation of the tracking control has been investigated in [20]. Linear MPC inherits the advantage of the quadratic programming formulation that can be efficiently solved by off-the-shelf numerical algorithms [21], which is promising in regard to real-time implementation. On the other hand, the disadvantage of this configuration is obviously drawn from the approximation error by the local linearization. Naturally, the nonlinear MPC formulation has also been exploited for the AUV tracking control problem [22] with the sacrifice of algorithm efficiency to model accuracy. Nevertheless, among the existing results of MPC tracking control, they either discuss at the velocity level, i.e., only take the kinematic equations as the system model, e.g., [22]–[26], or consider both kinematics and dynamics but no rigorous mathematical proof of closed-loop stability was conducted in the controller design, e.g., [20]. This motivates the main objective of this paper: To design a stable MPC tracking controller considering complete AUV dynamics and kinematics.

On the other hand, path planning plays an important role in the tracking control. The conventional marine vessel guidance system [7] that generates whole-journey way-points does not work for AUVs, because oftentimes the global oceanic information is not available *a priori*. Instead, fast, reactive, and dynamic path planning methods are desirable [27], [28]. Recently, a spline-based path planning method was reported in [29] for autonomous mining vehicle applications. In view of the similar essentials of autonomous vehicles, the basic principle is probably applicable to AUVs as well. The global environment information is assumed in [29]. Considering the practical operation of AUVs with a limited sensing range, the second objective of this paper appears: To accommodate the spline-based path planning method to the limited perceiving capability of AUVs in the tracking control application.

In this paper, by introducing a virtual reference system, the tracking control problem can be transformed to the regulation problem of the error dynamics. The optimal value function of the optimization problem solved at each sampling instant is shown nonincreasing along the trajectory of the closed-loop system, which guarantees the asymptotic stability of the closed-loop control system. Furthermore, a dynamic path planning method is developed based on the receding horizon optimizations, which enables the natural integration of the path planning the tracking control.

The contributions of this paper are three fold.

1) A novel receding horizon optimization formulation is proposed for the spline-based path planning method that accommodates the practically limited perceiving capability of AUVs.
2) An MPC-based tracking controller is designed with guaranteed closed-loop stability. To the best of our knowledge, it is the first time to explicitly conduct the stabilizing conditions of MPC-based trajectory tracking control for marine vehicles.
3) A unified receding horizon optimization scheme, seamlessly integrating the path planning and the MPC tracking control, is proposed for the AUV application.

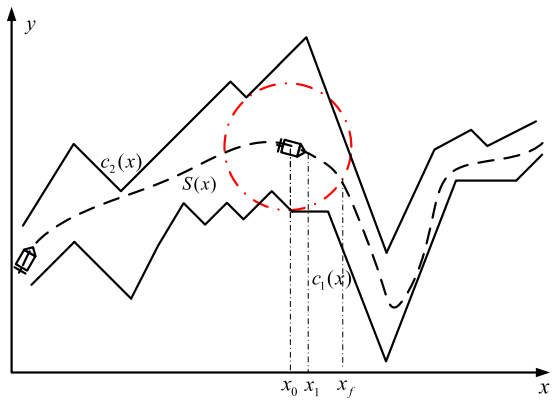


Fig. 1. Illustration of the AUV control problem.

The remainder of this paper is organized as follows. In Section II, the problem statement is presented. In Section III, we develop a receding horizon path planing method for the vehicle. Section IV-A introduces the mathematical model that is used for the AUV tracking control; the formulation of the tracking control problem into the MPC scheme is discussed in Section IV-B, together with the detailed controller design. In Section V, the overall control algorithm which combines the MPC and the path planning is depicted. Simulation results are demonstrated in Section VI. Section VII provides some conclusive remarks for the entire paper.

The notations are explained as follows. The column operation $[\rho_1^{\mathrm{T}}, \ldots, \rho_n^{\mathrm{T}}]^{\mathrm{T}}$ is denoted by $col(\rho_1, \ldots, \rho_n)$. For a function $f(x)$, the time derivative is denoted by $\dot{f}$ while the derivative with respect to x is denoted by f'. The diagonal operation is abbreviated by $\mathrm{diag}(\cdot)$.

II. Problem Statement

The combined problem of path planning and tracking control arises in many AUV applications such as marine source seeking [30] and deep-sea archaeological exploration [31]. As illustrated in Fig. 1, the workspace of AUVs is confined by two polygonal chains $c_1(x)$ and $c_2(x)$ that reflect the topography of the seafloor or the structure of an underwater shipwreck. A reference path $S(x)$ has to be planned for safety reasons.

Since the AUV sway velocity is considerably smaller than its surge counterpart, the nonholonomic kinematic equations can well approximate the vehicle motion [7] at the velocity level, i.e.,

$$\begin{aligned} \dot{x} &= u\cos\psi - v\sin\psi \approx u_0\cos\psi \\ \dot{y} &= u\sin\psi + v\cos\psi \approx u_0\sin\psi \\ \dot{\psi} &= r \end{aligned} \tag{1}$$

where $[x, y, \psi]^{\mathrm{T}}$ denote the position and orientation; $[u, v, r]^{\mathrm{T}}$ stand for the surge, sway, and yaw velocities while u_0 denotes the resultant tangential velocity of the vehicle.

To precisely track a desired parameterized path $p(s) = [x_R(s), y_R(s)]^{\mathrm{T}}$, the open-loop tangential velocity u_{oc} and yaw velocity r_{oc} can be calculated for the kinematic model (1) pro-

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vided no disturbances and initial errors

$$u_{\rm oc}(t) = \sqrt{x_R'^2 + y_R'^2}\,\dot{s} \tag{2a}$$

$$r_{\rm oc}(t) = \frac{x_R' y_R'' - y_R' x_R''}{x_R'^2 + y_R'^2}\dot{s} = u_{\rm oc}(t)\kappa(s) \tag{2b}$$

where $\kappa(s)$ represents the curvature along the path.

From (2), it is readily seen the smaller curvature makes the steering easier. In this sense, a reference path with minimum curvature is practically desirable for AUVs.

However, the effective sensing range is practically short in water (described as the dashed circle shown in Fig. 1). The reference path has to be dynamically generated as the vehicle moves forward, i.e., the path planning and tracking control are inherently coupled. Therefore, the AUV control task contains the following two aspects.

1) Dynamic Path Planning: Based on updated local information dynamically plan a minimum curvature reference path for the AUV.
2) Precise Tracking Control: Based on the planned reference path steer the AUV to converge and precisely track the path.

III. Minimum Curvature Receding Horizon Path Planning

Inspired by [32] and [29], we use the spline functions as the path template. A spline function represented in b-form is defined as follows:

$$S(x) = \sum_{i=1}^{n} \alpha_i S_{i,k}(x) \tag{3}$$

where n is the number of knots, α_i are the control parameters, and $S_{i,k}(x)$ are kth order basis splines defined recursively,

$$S_{i,1}(x) = \begin{cases} 1, & x_i \le x < x_{i+1} \\ 0, & \text{otherwise} \end{cases}$$

and

$$S_{i,k}(x) = \frac{(x - x_i)S_{i,k-1}(x)}{x_{i+k-1} - x_i} + \frac{(x_{i+k} - x)S_{i+1,k-1}(x)}{x_{i+k} - x_{i+1}}.$$

For a spline $S(x)$, the measure of smoothness is chosen as (4), which is the objective function in the reference path planning calculation

$$F(x) = \int_{s(x_0)}^{s(x_f)} \kappa'(s)^2 ds. \tag{4}$$

Considering the relationship between the arc-length s and the Cartesian coordinate x, the objective function in (4) is equivalent to

$$F(x) = \int_{x_0}^{x_f} \frac{\kappa'(x)^2}{\sqrt{1 + S'(x)^2}} dx$$

where $\kappa(x)$ represents the curvature of the spline

$$\kappa(x) = \frac{S''(x)}{(1 + S'(x)^2)^{3/2}}.$$

For simplicity, we assume that the yaw plane environment is restricted by two polygonal chains (see Fig. 1). Then the path planning problem can be formulated as the following optimization problem.

Problem 1: Given the restriction polygonal chains $c_1(x) \le c_2(x)$, with proper numbers of knots n, order o and end multiplicity l, compute the spline function $S(\alpha, x)$ defined on $\Omega = [x_0, x_f]$ such that

$$\begin{aligned} \min_{\alpha} \ & F(\alpha, x) \\ \text{s.t.} \ & c_1(x) \le S(\alpha, x) \le c_2(x) \end{aligned}$$

where $\alpha = [\alpha_1, \ldots, \alpha_n]^{\rm T}$. Here, we denote the spline as $S(\alpha, x)$ and objective function as $F(\alpha, x)$ to explicitly point out the dependence on the control parameters that are the decision variables in the optimization. In other places, when the spline is already determined, we simply use the notation $S(x)$ as per established convention [33]. Also it should be noticed that the spline template is always chosen of at least order three to guarantee the continuity of the acceleration.

Remark 1: It is worth noting that the number of control parameters has an explicit relation to the number of knots, thus the selection of knots appears a significant issue. Extensive study on knot placement can be found in [34].

For this application, the larger number of knots makes the description of the environment more accurate, but it also makes the calculation of reference path $S(x)$ more time consuming. In view of the limited computational resource and memory size on chip, usually the number of knots is required to be as small as possible without losing main characteristics of the environment. A possible way is to choose the local extrema of the environment and connect these extrema sequentially. A safe margin concept can be used [29].

Another issue regarding computational complexity is the inequality constraints $c_1(x) \le S(\alpha, x) \le c_2(x)$. The constraints are enforced numerically, i.e., we first sample these functions and then impose the inequality constraints at discrete points. Therefore, to increase the number of samples is to enlarge the size of the optimization problem. However, b-splines form a stable basis for splines and Runge's phenomenon in which oscillation may occur between knots can be avoided. In practice, a small number of samples is often sufficient.

Remark 2: Although we simplify the unknown yaw plane environment as the area bounded by two polygonal chains, more complicated environment can be tackled by transforming it into constraints of the optimization problem or equivalently feasible regions for the nonlinear programming (NLP) algorithm. NLP algorithms such as sequential quadratic programming (SQP) method and interior-point method [21] usually require a connected feasible region. However, complicated environment may be characterized as several disjoint feasible subregions. This will result in more than one time NLP to calculate the reference path.

Because the sensing radius is short in practice, the global optimization cannot be performed for the AUV path planning. Instead, the receding horizon scheme can help to approximate the minimum curvature reference path. This idea can be illustrated through the following example: Suppose the sensing range is 5 m, the dashed circle in Fig. 1; at a specific position, a local reference path, the dashed line segment between x_0 and x_f, is generated based on the current 5-m data; when the AUV moves 1 m forward to x_1, a new reference path, between the new start

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point $\bar{x}_0 = x_1$ and new end point $\bar{x}_f$, will be generated based on the new 5-m measurement data. In this case, the continuity between the neighbor splines has to be taken into consideration. Therefore, we impose the continuity conditions as constraints in the optimization formulation.

Problem 2: Given measured environment information $c_1(x) \le c_2(x)$ only defined on $\Omega = [x_0, x_f]$ and the last calculated reference path $S_0(x)$, with proper numbers of knots n, order o, and end multiplicity l, compute the spline function $S(\alpha, x)$ defined on Ω such that

$$\begin{aligned} \min_{\alpha} \ & F(\alpha, x) \\ \text{s.t.} \ & c_1(x) \le S(\alpha, x) \le c_2(x) \\ & \| S(\alpha, x_0) - S_0(x_0) \| \le \epsilon_1 \\ & \| S'(\alpha, x_0) - S_0'(x_0) \| \le \epsilon_2 \\ & \| S''(\alpha, x_0) - S_0''(x_0) \| \le \epsilon_3 \end{aligned}$$

where $\alpha = [\alpha_1, \ldots, \alpha_n]^{\mathrm{T}}$ and ϵ_i are the tolerances for discontinuity. We should notice that the start point of the current spline x_0 is coincident with the second knot of the previous calculated spline.

IV. Model Predictive Tracking Control

In this section, we first present the model of *Falcon* under investigation. Then, we propose the design of the MPC tracking control law so that the AUV can precisely track the reference trajectory generated in Section III.

A. AUV Modeling

For an AUV, the well established six degree of freedom dynamic equations can be expressed as follows [7]:

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \tag{5}$$

where $\mathbf{v}$ denotes the body-fixed velocity vector, and $\boldsymbol{\eta}$ denotes the earth-fixed position and orientation vector. $\mathbf{M}$ represents the inertia matrix including added mass, $\mathbf{C}(\mathbf{v})$ is the state-dependent matrix of Coriolis and centripetal terms, $\mathbf{D}(\mathbf{v})$ is the hydrodynamic damping and lift matrix, $\mathbf{g}(\boldsymbol{\eta})$ represents the vector of gravitational forces and moments, and $\boldsymbol{\tau}$ is the vector of thrust forces and moments.

In the yaw plane, we introduce three mild assumptions to simplify the mathematical derivation.

1) The AUV is with three planes of symmetry.
2) The mass distribution is homogeneous.
3) The pitch and roll motions are neglected.

Furthermore, the origin of the body-fixed reference frame is selected to be the center of gravity. As a result, for the yaw plane tracking control, the system matrices in (5) are simplified as $\mathbf{M} = \mathrm{diag}(M_x, M_y, M_\psi)$, $\mathbf{D}(\mathbf{v}) = \mathrm{diag}(X_u, Y_v, N_r) + \mathrm{diag}(D_u|u|, D_v|v|, D_r|r|)$, $\mathbf{g}(\boldsymbol{\eta}) = \mathbf{0}$ and

$$\mathbf{C}(\mathbf{v}) = \begin{bmatrix} 0 & 0 & -M_y v \\ 0 & 0 & M_x u \\ M_y v & -M_x u & 0 \end{bmatrix}$$

where $\mathbf{v} = [u, v, r]^{\mathrm{T}}$ denote the surge, sway, and yaw velocities, while $\boldsymbol{\eta} = [x, y, \psi]^{\mathrm{T}}$ represent the corresponding position and orientation; $\boldsymbol{\tau} = [F_u, F_v, F_r]^{\mathrm{T}}$ is the total force and moment exerting on the vehicle; $M_x = m - X_{\dot{u}}$, $M_y = m - Y_{\dot{v}}$ and $M_\psi = I_z - N_{\dot{r}}$ are the inertia terms including add mass; X_u, Y_v and N_r are linear drag terms, while D_u, D_v, and D_r are corresponding quadratic drag terms.

To further expand the dynamic equation (5) into an element-wise expression, we have

$$\begin{aligned} \dot{u} &= \frac{M_y}{M_x} vr - \frac{X_u}{M_x} u - \frac{D_u}{M_x} u|u| + \frac{F_u}{M_x} \\ \dot{v} &= -\frac{M_x}{M_y} ur - \frac{Y_v}{M_y} v - \frac{D_v}{M_y} v|v| + \frac{F_v}{M_y} \\ \dot{r} &= \frac{M_x - M_y}{M_\psi} uv - \frac{N_r}{M_\psi} r - \frac{D_r}{M_\psi} r|r| + \frac{F_r}{M_\psi}. \end{aligned} \tag{6}$$

On the other hand, the AUV kinematic equations that translates the body-fixed velocity into the earth-fixed velocity expression, can be written as follows:

$$\dot{\boldsymbol{\eta}} = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} \triangleq \mathbf{J}(\boldsymbol{\eta})\mathbf{v}. \tag{7}$$

Further expanding the kinematic equation (7) into an element-wise expression, we have

$$\begin{aligned} \dot{x} &= u\cos\psi - v\sin\psi \\ \dot{y} &= u\sin\psi + v\cos\psi \\ \dot{\psi} &= r. \end{aligned} \tag{8}$$

We define the system state $\mathbf{x} \triangleq col(\boldsymbol{\eta}, \mathbf{v})$ and view $\boldsymbol{\tau}$ as the generalized control input, from (5) and (7), we have the general expression of the AUV model

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{J}(\boldsymbol{\eta})\mathbf{v} \\ \mathbf{M}^{-1}(\boldsymbol{\tau} - \mathbf{C}\mathbf{v} - \mathbf{D}\mathbf{v} - \mathbf{g}) \end{bmatrix} \triangleq \bar{\mathbf{f}}(\mathbf{x}, \boldsymbol{\tau}).$$

Specifically, the *Falcon* [35] is equipped with four thrusters. The generalized control input is hence expressed as $\boldsymbol{\tau} = \mathbf{B}\mathbf{u}$, where $\mathbf{B}$ is the control matrix. $\mathbf{u} \triangleq [u_1, u_2, u_3, u_4]^{\mathrm{T}}$ representing the thrust from front port, rear starboard, front starboard, and rear port thrusters are the real control inputs. The overall control system model is built as follows:

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{J}(\boldsymbol{\eta})\mathbf{v} \\ \mathbf{M}^{-1}(\mathbf{B}\mathbf{u} - \mathbf{C}\mathbf{v} - \mathbf{D}\mathbf{v}) \end{bmatrix} \triangleq \mathbf{f}(\mathbf{x}, \mathbf{u}). \tag{9}$$

From a control point of view, the tracking control is challenging due to the nonlinearity and coupled dynamics in the model (9). Since the vehicle surge velocity does not keep constant for trajectory tracking tasks, the assumptions for model linearization cannot hold. Linear control techniques appear inappropriate to apply. In this regard, nonlinear MPC, which is capable of dealing with complex nonlinearity of the dynamics, seems an attractive option.

B. Stabilizing MPC Tracking Control

The reference path $S(x)$ can be viewed as the state trajectory $\boldsymbol{\eta}_R = [x_R, y_R, \psi_R]^{\mathrm{T}}$ generated by a reference system that owns

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the same kinematic equations of motion as the real vehicle's

$$\begin{aligned}\dot{x}_R &= u_R\cos\psi_R - v_R\sin\psi_R\\ \dot{y}_R &= u_R\sin\psi_R + v_R\cos\psi_R\\ \dot{\psi}_R &= r_R.\end{aligned} \tag{10}$$

Note that the planned spline $S(x)$ in Section III is not yet time parameterized, i.e., a trajectory. To make the reference spline a trajectory, an ancillary design of timing law is required. See Remark 3.

Decomposing the kinematic error $\boldsymbol{\eta}_e$ in the vessel parallel reference frame [7], [36], we have

$$\boldsymbol{\eta}_e = \begin{bmatrix} x_e\\ y_e\\ \psi_e\end{bmatrix} = \begin{bmatrix}\cos\psi & \sin\psi & 0\\ -\sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix} x_R - x\\ y_R - y\\ \psi_R - \psi\end{bmatrix}. \tag{11}$$

We further partition the velocity $\mathbf{v} = \mathbf{v}_f + \mathbf{v}_e$, where $\mathbf{v}_f = [u_R\cos\psi_e - v_R\sin\psi_e, u_R\sin\psi_e + v_R\cos\psi_e, r_R]^{\mathrm{T}}$ can be regarded as a feed-forward control action, and $\mathbf{v}_e = [u_e, v_e, r_e]^{\mathrm{T}}$ represents the feedback control action.

Differentiating (11) and substituting (8) and (10), we derive the kinematic error equations

$$\begin{aligned}\dot{x}_e &= y_e r_R - u_e + y_e r_e\\ \dot{y}_e &= -x_e r_R - v_e - x_e r_e\\ \dot{\psi}_e &= -r_e.\end{aligned} \tag{12}$$

Analogously, we view $\mathbf{v}_R = [u_R, v_R, r_R]^{\mathrm{T}}$ as the state trajectories of the reference system that owns the same dynamic properties as the vehicle

$$\begin{aligned}\dot{u}_R &= \frac{M_y}{M_x}v_R r_R - \frac{X_u}{M_x}u_R - \frac{D_u}{M_x}u_R|u_R| + \frac{F_{\mathrm{uR}}}{M_x}\\ \dot{v}_R &= -\frac{M_x}{M_y}u_R r_R - \frac{Y_v}{M_y}v_R - \frac{D_v}{M_y}v_R|v_R| + \frac{F_{vR}}{M_y}\\ \dot{r}_R &= \frac{M_x - M_y}{M_\psi}u_R v_R - \frac{N_r}{M_\psi}r_R - \frac{D_r}{M_\psi}r_R|r_R| + \frac{F_{rR}}{M_\psi}.\end{aligned} \tag{13}$$

Differentiating both sides of $\mathbf{v}_e = \mathbf{v} - \mathbf{v}_f$ and applying the feedback linearization technique [14], we derive the overall error dynamics for the AUV tracking control problem

$$\dot{\mathbf{x}}_e = \begin{bmatrix}\dot{x}_e\\ \dot{y}_e\\ \dot{\psi}_e\\ \dot{u}_e\\ \dot{v}_e\\ \dot{r}_e\end{bmatrix} = \begin{bmatrix} y_e r_R - u_e + y_e r_e\\ -x_e r_R - v_e - x_e r_e\\ -r_e\\ \tau_u\\ \tau_v\\ \tau_r\end{bmatrix} \triangleq \mathbf{f}_e(\mathbf{x}_e, \boldsymbol{\tau}_e) \tag{14}$$

where $\boldsymbol{\tau}_e = [\tau_u, \tau_v, \tau_r]^{\mathrm{T}}$. Detailed derivations can be found in Appendix A.

Obviously, $(\mathbf{0}, \mathbf{0})$ is an equilibrium point of (14). Comparing (13) and (28)–(30) at $(\mathbf{0}, \mathbf{0})$, we find that $F_u = F_{\mathrm{uR}}, F_v = F_{vR}$, and $F_r = F_{rR}$, which means that the tracking of the reference path is equivalent to stabilizing the error system (14) at the equilibrium point.

To stabilize the error system, the MPC technique is used. The stabilization is realized through the minimization of the following performance index:

$$J(t, \mathbf{x}_e(t), \boldsymbol{\tau}_e(t)) = h(\mathbf{x}_e(T)) + \int_t^T \ell(\mathbf{x}_e(s), \boldsymbol{\tau}_e(s))ds \tag{15}$$

where the initial time is viewed as 0, $h(\cdot)$ is the terminal state penalty satisfying $h(\mathbf{0}) = 0$ and $h(\mathbf{x}_e) > 0$ for any $\mathbf{x}_e \neq \mathbf{0}$; $\ell(\mathbf{x}_e, \boldsymbol{\tau}_e) = \mathbf{x}_e^{\mathrm{T}}\mathbf{Q}\mathbf{x}_e + \boldsymbol{\tau}_e^{\mathrm{T}}\mathbf{R}\boldsymbol{\tau}_e$ is the stage cost with $\mathbf{Q} > 0$, $\mathbf{R} \geq 0$. Then the optimal control problem to be solved at each sampling time instant is constructed as follows.

Problem 3: Given the current error state $\mathbf{x}_e$, compute the optimal control force $\boldsymbol{\tau}_e(t)$ by solving the following optimization problem:

$$\begin{aligned}\min_{\boldsymbol{\tau}_e}.\ & J(t, \mathbf{x}_e(t), \boldsymbol{\tau}_e(t))\\ \text{s.t.}\ & \dot{\mathbf{x}}_e(t) = \mathbf{f}_e(\mathbf{x}_e(t), \boldsymbol{\tau}_e(t)),\ \text{for } t \in [0, T]\\ & \boldsymbol{\tau}_e(t) \in T_e,\ \text{for } t \in [0, T]\\ & \mathbf{x}_e(T) \in X_f\end{aligned} \tag{16}$$

where T_e is assumed to be a compact set containing the origin in the interior and X_f the terminal constraint set.

The MPC algorithm for the AUV tracking control problem can be briefly described as follows.

1) The optimization problem (16) is solved with the current error state $\mathbf{x}_e(t_0)$ as the initial condition. Let $\bar{\boldsymbol{\tau}}_e(t)$ denote the solution.
2) The system only uses $\bar{\boldsymbol{\tau}}_e(t)$ to calculate $\boldsymbol{\tau}$ via (28)–(30) for one sampling period: $\boldsymbol{\tau}_e(t) = \bar{\boldsymbol{\tau}}_e(t)$ for $[t_0, t_0 + \delta]$.
3) At time $t_0 + \delta$, the optimization problem (16) is solved again using the latest measured data $\mathbf{x}_e(t_0 + \delta)$.

The above procedure will repeat until accomplishing the AUV tracking task. However, it is well known that optimality does not automatically guarantee the closed-loop stability due to the finite prediction horizon. To solve this problem, we need to appropriately design the local controller $\boldsymbol{\tau}_e^L$, the terminal state penalty $h(\cdot)$ and the terminal state constraint X_f [37].

Theorem 1: For nonlinear systems like (14), the closed-loop system controlled by the MPC algorithm is asymptotically stable if the following conditions are satisfied.

C1: $\mathbf{0} \in X_f$.

C2: There exists a local controller $\boldsymbol{\tau}_e^L(t)$ satisfying $\dot{h}(\mathbf{x}_e) + \ell(\mathbf{x}_e, \boldsymbol{\tau}_e^L(t)) \leq 0$ for all $\mathbf{x}_e \in X_f$.

C3: Let $\bar{\boldsymbol{\tau}}_e(t|t_0)$ denote the solution of the optimization problem (16) at time t_0, the initial guess for the optimization at the next sampling time instant $t_0 + \delta$ is chosen

$$\hat{\boldsymbol{\tau}}_e(t|t_0 + \delta) = \begin{cases}\bar{\boldsymbol{\tau}}_e(t|t_0)\ \text{ for }\ t_0 + \delta \leq t \leq t_0 + T\\ \boldsymbol{\tau}_e^L(t)\ \text{ for }\ t_0 + T \leq t \leq t_0 + T + \delta.\end{cases} \tag{17}$$

Proof. See Appendix B. ∎

Now, we design the MPC tracking controller based on *Theorem 1*: First, we select the initial guess using the way of C3. Second, we select the local controller to be a linear state feedback $\boldsymbol{\tau}_e^L(t) = \mathrm{K}\mathbf{x}_e$. Third, we set the terminal state penalty $h(\mathbf{x}_e) \triangleq \frac{1}{2}\mathbf{x}_e^{\mathrm{T}}\mathbf{x}_e$. Specifically, let $\mathrm{K} \triangleq [\mathrm{K}_1\ \mathrm{K}_2]$ where

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$\mathrm{K}_1 = \mathrm{diag}(\alpha_1, \beta_1, \gamma_1)$ and $\mathrm{K}_2 = -\mathrm{diag}(\alpha_2, \beta_2, \gamma_2)$ with $\alpha_i > 0, \beta_i > 0,\ \gamma_i > 0$; $\mathbf{Q} = \mathrm{diag}(\mathrm{q}_{11}, \mathrm{q}_{22}, \mathrm{q}_{33}, \mathrm{q}_{44}, \mathrm{q}_{55}, \mathrm{q}_{66})$ and $\mathbf{R} = \mathrm{diag}(\mathrm{r}_{11}, \mathrm{r}_{22}, \mathrm{r}_{33})$ with $q_{\mathrm{ii}} > 0, r_{\mathrm{ii}} \geq 0$, then we calculate the following:

$$\begin{aligned}\dot{h}(\mathbf{x}_e) =&\ \dot{x}_e x_e + \dot{y}_e y_e + \dot{\psi}_e \psi_e + \dot{u}_e u_e + \dot{v}_e v_e + \dot{r}_e r_e \\ =&\ (y_e r_R - u_e + y_e r_e) x_e + (-x_e r_R - v_e - x_e r_e) y_e \\ &- \psi_e r_e + u_e \tau_u^L + v_e \tau_v^L + r_e \tau_r^L \\ =&\ -x_e u_e - y_e v_e - \psi_e r_e + u_e \tau_u^L + v_e \tau_v^L + r_e \tau_r^L\end{aligned}$$

and

$$\begin{aligned}\ell(\mathbf{x}_e, \boldsymbol{\tau}_e^L(t)) =&\ q_{11} x_e^2 + q_{22} y_e^2 + q_{33} \psi_e^2 + q_{44} u_e^2 + q_{55} v_e^2 \\ &+ q_{66} r_e^2 + r_{11} (\tau_u^L)^2 + r_{22} (\tau_v^L)^2 + r_{33} (\tau_r^L)^2.\end{aligned}$$

Substituting $\boldsymbol{\tau}_e^L(t) = \mathrm{K}\mathbf{x}_e$, we have

$$\begin{aligned}&\dot{h}(\mathbf{x}_e) + \ell(\mathbf{x}_e, \boldsymbol{\tau}_e^L(t)) \\ &= -x_e u_e - v_e y_e - \psi_e r_e + \alpha_1 x_e u_e - \alpha_2 u_e^2 + \beta_1 y_e v_e \\ &- \beta_2 v_e^2 + \gamma_1 \psi_e r_e - \gamma_2 r_e^2 + q_{11} x_e^2 + q_{22} y_e^2 + q_{33} \psi_e^2 \\ &+ q_{44} u_e^2 + q_{55} v_e^2 + q_{66} r_e^2 + r_{11} (\alpha_1 x_e - \alpha_2 u_e)^2 \\ &+ r_{22} (\beta_1 y_e - \beta_2 v_e)^2 + r_{33} (\gamma_1 \psi_e - \gamma_2 r_e)^2 \\ &= (q_{44} - \alpha_2 + r_{11} \alpha_2^2) u_e^2 + (q_{55} - \beta_2 + r_{22} \beta_2^2) v_e^2 \\ &+ (q_{66} - \gamma_2 + r_{33} \gamma_2^2) r_e^2 + (q_{11} + r_{11} \alpha_1^2) x_e^2 \\ &+ (\alpha_1 - 1 - 2\alpha_1 \alpha_2 r_{11}) x_e u_e + (q_{22} + r_{22} \beta_1^2) y_e^2 \\ &+ (\beta_1 - 1 - 2\beta_1 \beta_2 r_{22}) y_e v_e + (q_{33} + r_{33} \gamma_1^2) \psi_e^2 \\ &+ (\gamma_1 - 1 - 2\gamma_1 \gamma_2 r_{33}) \psi_e r_e.\end{aligned}$$

Then we select these parameters satisfying

$$\begin{aligned}q_{11} + r_{11} \alpha_1^2 + \alpha_1 - 1 - 2\alpha_1 \alpha_2 r_{11} \leq&\ 0 \\ q_{22} + r_{22} \beta_1^2 + \beta_1 - 1 - 2\beta_1 \beta_2 r_{22} \leq&\ 0 \\ q_{33} + r_{33} \gamma_1^2 + \gamma_1 - 1 - 2\gamma_1 \gamma_2 r_{33} \leq&\ 0 \\ q_{44} - \alpha_2 + r_{11} \alpha_2^2 \leq&\ 0 \\ q_{55} - \beta_2 + r_{22} \beta_2^2 \leq&\ 0 \\ q_{66} - \gamma_2 + r_{33} \gamma_2^2 \leq&\ 0\end{aligned} \tag{18}$$

and the terminal constraint satisfying

$$\begin{aligned}|u_e| \geq&\ |x_e|,\ x_e u_e \geq 0 \\ |v_e| \geq&\ |y_e|,\ y_e v_e \geq 0 \\ |r_e| \geq&\ |\psi_e|,\ \psi_e r_e \geq 0.\end{aligned} \tag{19}$$

We further choose the terminal constraint set to be the sublevel set of $h(\cdot)$, i.e.,

$$X_{f1} = \left\{ \mathbf{x}_e \in \mathbb{R}^6 \mid \frac{1}{2} \mathbf{x}_e^{\mathrm{T}} \mathbf{x}_e \leq \sigma \right\}. \tag{20}$$

As a result, the condition C2 is satisfied. Define $X_f = X_{f1} \cap X_{f2}$, where

$$X_{f2} = \{ \mathbf{x}_e \in \mathbb{R}^6 \mid (19) \}. \tag{21}$$

$C1$ can be satisfied in X_f.

Note that the conditions (18) are easy to satisfy. In fact, these parameters can be divided into three independent groups: $(q_{11}, q_{44}, r_{11}, \alpha_1, \alpha_2)$, $(q_{22}, q_{55}, r_{22}, \beta_1, \beta_2)$, and $(q_{33}, q_{66}, r_{33}, \gamma_1, \gamma_2)$. The selection among these groups are not coupled. Then for each group only two inequality conditions need to be satisfied.

Remark 3: Back to (10) and (13), we shall notice that the reference system plays a crucial role in the MPC tracking control. Generally speaking, finding such a reference system is nontrivial, and it might be even more difficult than the stabilization of the error system itself. Fortunately, for the AUV tracking problem the reference pair $(\mathbf{x}_R, \boldsymbol{\tau}_R)$ can be derived directly from the reference path $S(x)$. In particular, one possible choice can be determined in the following way.

1) Make the calculated spline $S(x)$ a reference trajectory of the kinematic state $\boldsymbol{\eta}_R$ by employing a predetermined timing law:
$$\begin{aligned}x_R =&\ \rho\, t \\ y_R =&\ S(x_R) \\ \psi_R =&\ \mathrm{atan2}(\dot{y}_R, \dot{x}_R)\end{aligned}$$
where $\rho > 0$ and atan2 is the four-quadrant inverse tangent operation.
2) Choose the reference velocity $\mathbf{v}_R$ to be $u_R = u_{\mathrm{oc}}, v_R = 0$, and $r_R = r_{\mathrm{oc}}$.
3) Calculate the reference control input $\boldsymbol{\tau}_R$ through (13).

Remark 4: For the AUV tracking control, the control constraints are given as follows:

$$\begin{aligned}F_{u,\min} \leq&\ F_u \leq F_{u,\max} \\ F_{\mathrm{v},\min} \leq&\ F_v \leq F_{v,\max} \\ F_{\mathrm{r},\min} \leq&\ F_r \leq F_{r,\max}.\end{aligned} \tag{22}$$

Then we can derive the error control constraints as

$$\begin{aligned}\frac{F_{u,\min} - F_1}{M_x} \leq&\ \tau_u \leq \frac{F_{u,\max} - F_1}{M_x} \\ \frac{F_{\mathrm{v},\min} - F_2}{M_y} \leq&\ \tau_v \leq \frac{F_{v,\max} - F_2}{M_y} \\ \frac{F_{\mathrm{r},\min} - F_3}{M_\psi} \leq&\ \tau_r \leq \frac{F_{r,\max} - F_3}{M_\psi}.\end{aligned} \tag{23}$$

Here, $F_1 = F_u - M_x \tau_u$, $F_2 = F_v - M_y \tau_v$, and $F_3 = F_r - M_\psi \tau_r$ that are explicitly expressed in (28)–(30).

Although (23) is time varying with the reference system, it is not necessarily explicitly calculated. Problem 3 is defined to facilitate the description of the stability theorem. In the MPC implementation, we can directly use the polyhedral input constraint (22) to calculate $\boldsymbol{\tau}$ but construct the cost function (15)

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using the elaborately defined error state $\mathbf{x}_e$ and control $\boldsymbol{\tau}_e$. Then the equivalent $\boldsymbol{\tau}_e = \boldsymbol{\tau} - F_F$ with $F_F = [F_1, F_2, F_3]^{\mathrm{T}}$.

Notice that the predetermined timing law may be carefully designed cf., [38] and [39] such that the reference system chosen in Remark 3 fits the actual system. Specifically, when the vehicle is moving along the reference spline $S(x)$, the reference control input can be expressed in an explicit manner

$$F_{\mathrm{uR}} = M_x \dot{u}_R + X_u u_R + D_u u_R |u_R| \tag{24a}$$

$$F_{vR} = M_x u_R r_R \tag{24b}$$

$$F_{rR} = M_\psi \dot{r}_R + N_r r_R + D_r r_R |r_R| \tag{24c}$$

where $\dot{u}_R$ and $\dot{r}_R$ can also be explicitly calculated by

$$\dot{u}_R = S'(x_R)S''(x_R)(1+S'(x_R)^2)^{-\frac{1}{2}}\dot{x}_R^2 + (1+S'(x_R)^2)^{\frac{1}{2}}\ddot{x}_R \tag{25a}$$

$$\dot{r}_R = \frac{(1+S'(x_R)^2)S'''(x_R) - 2S'(x_R)S''(x_R)^2}{(1+S'(x_R)^2)^2}\dot{x}_R^2 + \frac{S''(x_R)}{(1+S'(x_R)^2)}\ddot{x}_R. \tag{25b}$$

Together with the reference state defined in Remark 3, we observe that one can always adjust x_R, i.e., the value of ρ to satisfy the input constraints. In the simulation, we use a fixed ρ as a simple example.

In addition, for the specific platform *Falcon*, the calculated generalized forces and moments $\boldsymbol{\tau} = [F_u, F_v, F_r]^{\mathrm{T}}$ will be transformed into real thrusts by $\mathbf{u} = \mathbf{B}^+\boldsymbol{\tau}$, where $\mathbf{B}^+$ is the Moore–Penrose pseudoinverse.

V. Integrated Receding Horizon Path Planning and Control: Algorithm Description

In view of MPC's sampled data nature, the tracking control can be conveniently integrated with the proposed receding horizon path planning method.

Let $\mathcal{D}_i$ denote the measured data for local environment information. The dimension of sampled reference path $S(x)$ between the first and second knot is $\mathcal{M}$. And the prediction horizon $T = \mathcal{N}\delta$ with $\mathcal{N} < \mathcal{M}$. Based on Problems 1–3, the integrated path planning and tracking control algorithm is summarized in Algorithm 1.

In Algorithm 1 two things need to be clarified. First, within the *for* iterations, we simply use $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ to denote the control process that to apply the calculated control signal for one sampling period and to measure the system state at next sampling instant. Second, since the prediction horizon $\mathcal{N}$ is fixed and the path planning is receding horizon, when the first segment of the current spline between the first knot and second knot cannot provide enough samples of reference state $\mathbf{x}_r(k)$ from $k = 1$ to $\mathcal{N}$, i.e., $\mathcal{M} - k < \mathcal{N}$, another round of path planning needs to be performed starting from the second knot of the current spline.

The block diagram that represents the integrated control scheme consisting of the path planner and the MPC tracking controller, is depicted in Fig. 2.

Algorithm 1:

$i = 0$.
Let $\mathbf{x}_0$ denote the initial state of the AUV.
Given sensor data $\mathcal{D}_i$, calculate the reference path by solving *Problem 1*. Let $S_i(x)$ be the solution.
procedure
    $k = 0$.
    for $k \leq \mathcal{M} - \mathcal{N}$ **do**
        Given $S_i(x)$ calculate the reference pair $(\mathbf{x}_R, \boldsymbol{\tau}_R)$ according to *Remark 3*; solve *Problem 3* and use (28)–(30) to calculate $\bar{\boldsymbol{\tau}}(t)$; for $k\delta \leq t \leq (k+1)\delta$, $\mathbf{u} = \mathbf{B}^+\bar{\boldsymbol{\tau}}$; $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$.
        $k = k + 1$.
    end for
    $i = i + 1$.
    Collect another set of sensor data $\mathcal{D}_i$, calculate reference path by solving *Problem 2*. Let $S_i(x)$ be the solution.
    for $k \leq \mathcal{M}$ **do**
        Calculate $(\mathbf{x}_R, \boldsymbol{\tau}_R)$ using both $S_{i-1}(x)$ and $S_i(x)$; calculate $\bar{\boldsymbol{\tau}}(t)$; for $k\delta \leq t \leq (k+1)\delta$, $\mathbf{u} = \mathbf{B}^+\bar{\boldsymbol{\tau}}$; $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, let $\mathbf{x}_f$ denote the system state at time $(k+1)\delta$.
        $k = k + 1$.
    end for
end procedure
$\mathbf{x}_0 = \mathbf{x}_f$, repeat **procedure**.

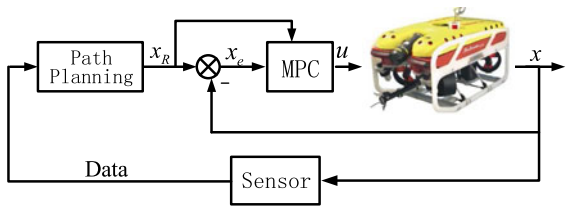


Fig. 2. Control block diagram.

With the aid of block diagram, we can explain the *procedure* in Algorithm 1 as follows:

1) The local environment $\mathcal{D}_i$ is detected. The mathematical representation of the workspace (c_1, c_2) is calculated according to current state $\mathbf{x}$ of the AUV, based on which the path planner constructs a receding horizon optimization problem, i.e., Problem 2 to determine the minimum curvature spline path $S_i(x)$.
2) The path $S_i(x)$ is then augmented and viewed as the trajectory of a virtual reference system that provides the reference state $(\mathbf{x}_R, \boldsymbol{\tau}_R)$ for the MPC tracking controller. A feed-forward channel exists to calculate the control effort F_F corresponding to the compensation of the time-varying portion of the reference path. Based on an elaborately defined error dynamics $(\mathbf{x}_e, \boldsymbol{\tau}_e)$, the receding horizon optimization problem, i.e., Problem 3, is

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constructed and then solved to obtain the tracking control signal $\mathbf{u} = \mathbf{B}^{+}\bar{\boldsymbol{\tau}}$.

3) As the AUV moves, the newly measured data will be collected and used to generate an updated reference path. Repeat from 1).

Remark 5: For the MPC tracking control, it is worth noting that at each sampling instant, only $\mathcal{N}$ steps of the reference states are used to construct the optimization problem. According to Algorithm 1, we have $\mathcal{N} < \mathcal{M}$ and every time (except for the first time) the reference path is planned starting from $\mathcal{N}$ steps ahead of the vehicle position. In this case, the reference path is equivalent to a predefined one although it is actually generated as the vehicle moves forward. The closed-loop stability of the tracking control will not be affected by the integration of the receding horizon path planning.

VI. Simulation Results

A. Parameter Selection

For the path planning, it is assumed that the vehicle can sense the environment up to 5 m ahead; we uniformly distribute 6 knots on the 5-m range and insert four samples between knots; choose $o = 4$, $l = 4$ and update the reference path with current 5-m data whenever the vehicle moves 1 m forward in the x-axis. $\epsilon_1 = \epsilon_2 = \epsilon_3 = 10^{-2}$ (m).

For the AUV, we extract the parameters from the identified dynamic model of *Falcon* [40]. The mass $m = 116$ (kg); the moment of inertia with respect to z-axis $I_z = 13.1$ (kg·m^2). The hydrodynamic coefficients $X_{\dot{u}} = -167.6$, $Y_{\dot{v}} = -477.2$, $N_{\dot{r}} = -15.9$, $X_u = 26.9$, $Y_v = 35.8$, $N_r = 3.5$, $D_u = 241.3$, $D_v = 503.8$, and $D_r = 76.9$. $F_{u,\max} = -F_{u,\min} = 2000$ (N), $F_{v,\max} = -F_{\mathrm{v,min}} = 2000$ (N), and $F_{r,\max} = -F_{\mathrm{r,min}} = 1000$ (Nm). The control matrix

$$\mathbf{B} = \begin{bmatrix} 0.7974 & 0.8643 & 0.8127 & 0.8270 \\ 0.6032 & 0.5029 & -0.5824 & -0.5610 \\ 0.2945 & -0.3302 & -0.2847 & 0.3505 \end{bmatrix}.$$

For MPC parameters, we select $\rho = 1$, $\delta = 0.1$ s, prediction horizon $T = 8\delta$. The reference system is chosen according to *Remark 3*. $\mathbf{Q} = \mathrm{diag}(\mathrm{q}_{11}, \mathrm{q}_{22}, \mathrm{q}_{33}, \mathrm{q}_{44}, \mathrm{q}_{55}, \mathrm{q}_{66}) = 0.4\ \mathbf{I}_6$, $\mathbf{R} = \mathrm{diag}(\mathrm{r}_{11}, \mathrm{r}_{22}, \mathrm{r}_{33}) = 0.01\mathbf{I}_3$, $\mathrm{K}_1 = \mathrm{diag}(\alpha_1, \beta_1, \gamma_1) = 0.5\mathbf{I}_3$, and $\mathrm{K}_2 = -\mathrm{diag}(\alpha_2, \beta_2, \gamma_2) = -\mathbf{I}_3$.

B. Tracking Performance

In the simulations, the optimization Problems 1–3 are solved by the embedded SQP algorithm in the MATLAB function *fmincon*.

The simulation results of the combined path planning and tracking control is shown in Fig. 3.

It can be observed that the generated reference path is smooth in light of small curvature that facilitates the tracking of AUVs. The MPC controller successfully steers the AUV asymptotically convergent to and then track the reference path in the presence of initial error. The average solution time per MPC step of is 1.9642 s that needs to be improved for real-time implementation. But this is obviously related to the computational capability of the central processing unit that goes beyond the scope of this paper. Accommodation of appropriate fast nonlinear MPC algorithms may also help. See the excellent review paper [41] for possible solutions.

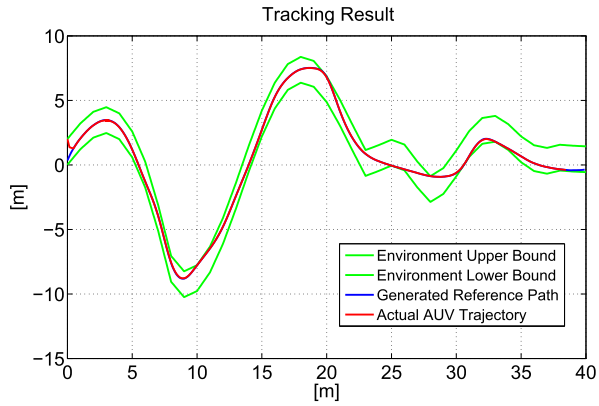


Fig. 3. Simulation results of the combined AUV problem.

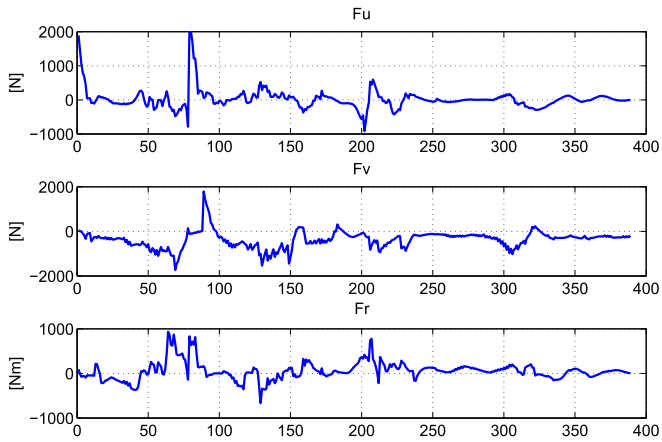


Fig. 4. Generalized control forces and moments.

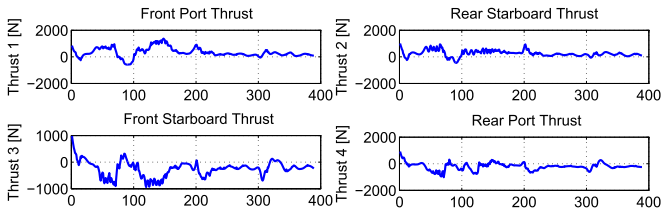


Fig. 5. Real thrust forces.

Fig. 4 shows the generalized control forces and moments, while Fig. 5 depicts the transformed real thrust forces. All the control signals take values within the permitted range. Here, we shall notice that the control inputs are not optimal in terms of fuel consumption (minimum control force) due to the selection of the reference system. As mentioned earlier, unlike the mobile robot whose sway velocity v is eliminated by friction, in water, the AUV will always slip sideways when take turns. Therefore, although we are able to conveniently find the reference system as stated in Remark 3, the AUV has to compensate the sway velocity v all the time, which increases the control

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effort. Nevertheless, if we view this point via another perspective, the compensation of v poses a request for the minimum curvature reference path, which is consistent with the proposed path planning method.

VII. Conclusion

We have presented, in this paper, a unified receding horizon optimization scheme to solve the combined path planning and tracking control problem for an AUV. A novel receding horizon path planning method was proposed with the assistance of spline path templates. Nonlinear MPC technique was employed to the AUV when tracking the planned spline path. Sufficient conditions for closed-loop stability were exploited, and further applied to guide the controller design. Sequentially, a control procedure inherently combining MPC and the proposed path planning method was developed. Simulation results using a *Falcon* dynamic model with realistic experimentally identified parameters revealed the effectiveness of the proposed control algorithm.

In this paper, we presume the disturbance is small for the AUV tracking application in constrained workspace, e.g., inside an underwater shipwreck. But it remains to analyze the robustness of the closed-loop system and characterize its disturbance rejection performance. Possibly an input-to-state stability extension [42] can be made to explicitly tolerate significant disturbances. Also improvements to the efficiency of the numerical algorithm to solve the NLP problems should be pursued. A fast nonlinear MPC algorithm cf.,[41] remains to be developed for real-time implementation.

Appendix A
Derivation of the Error Dynamics

Differentiating both sides of $\mathbf{v}_e = \mathbf{v} - \mathbf{v}_f$ yields

$$\begin{aligned}
\dot{u}_e &= \dot{u} - \dot{u}_R\cos\psi_e - u_R\sin\psi_e r_e + \dot{v}_R\sin\psi_e \\
&\quad - v_R\cos\psi_e r_e \\
\dot{v}_e &= \dot{v} - \dot{u}_R\sin\psi_e + u_R\cos\psi_e r_e - \dot{v}_R\cos\psi_e \\
&\quad - v_R\sin\psi_e r_e \\
\dot{r}_e &= \dot{r} - \dot{r}_R. \qquad (26)
\end{aligned}$$

Substituting (6) into (26), we have

$$\begin{aligned}
\dot{u}_e &= \frac{M_y}{M_x}(v_e + u_R\sin\psi_e + v_R\cos\psi_e)(r_e + r_R) \\
&\quad - \frac{X_u}{M_x}(u_e + u_R\cos\psi_e - v_R\sin\psi_e) - \frac{D_u}{M_x} \\
&\quad \times(u_e + u_R\cos\psi_e - v_R\sin\psi_e)|u_e + u_R\cos\psi_e \\
&\quad - v_R\sin\psi_e| - \dot{u}_R\cos\psi_e - u_R\sin\psi_e r_e + \dot{v}_R\sin\psi_e \\
&\quad - v_R\cos\psi_e r_e + \frac{F_u}{M_x} \\
\dot{v}_e &= -\frac{M_x}{M_y}(u_e + u_R\cos\psi_e - v_R\sin\psi_e)(r_e + r_R) \\
&\quad - \frac{Y_v}{M_y}(v_e + u_R\sin\psi_e + v_R\cos\psi_e) - \frac{D_v}{M_y} \\
&\quad \times(v_e + u_R\sin\psi_e + v_R\cos\psi_e)|v_e + u_R\sin\psi_e \\
&\quad + v_R\cos\psi_e| - \dot{u}_R\sin\psi_e + u_R\cos\psi_e r_e - \dot{v}_R\cos\psi_e \\
&\quad - v_R\sin\psi_e r_e + \frac{F_v}{M_y} \\
\dot{r}_e &= \frac{M_x - M_y}{M_\psi}(u_e + u_R\cos\psi_e - v_R\sin\psi_e)(v_e + u_R\sin\psi_e \\
&\quad + v_R\cos\psi_e) - \frac{N_r}{M_\psi}(r_e + r_R) - \frac{D_r}{M_\psi}(r_e + r_R)|r_e \\
&\quad + r_R| - \dot{r}_R + \frac{F_r}{M_\psi}. \qquad (27)
\end{aligned}$$

Apply the linear feedback technique [14] and design F_u, F_v, and F_r as follows:

$$\begin{aligned}
F_u &= -M_y(v_e + u_R\sin\psi_e + v_R\cos\psi_e)(r_e + r_R) + X_u(u_e \\
&\quad + u_R\cos\psi_e - v_R\sin\psi_e) + D_u(u_e + u_R\cos\psi_e \\
&\quad - v_R\sin\psi_e)|u_e + u_R\cos\psi_e - v_R\sin\psi_e| + M_x(\dot{u}_R\cos\psi_e \\
&\quad + u_R\sin\psi_e r_e - \dot{v}_R\sin\psi_e + v_R\cos\psi_e r_e) + M_x\tau_u \qquad (28) \\
F_v &= M_x(u_e + u_R\cos\psi_e - v_R\sin\psi_e)(r_e + r_R) \\
&\quad + Y_v(v_e + u_R\sin\psi_e + v_R\cos\psi_e) + D_v(v_e + u_R\sin\psi_e \\
&\quad + v_R\cos\psi_e)|v_e + u_R\sin\psi_e + v_R\cos\psi_e| + M_y(\dot{u}_R\sin\psi_e \\
&\quad - u_R\cos\psi_e r_e + \dot{v}_R\cos\psi_e + v_R\sin\psi_e r_e) + M_y\tau_v \qquad (29) \\
F_r &= (M_y - M_x)(u_e + u_R\cos\psi_e - v_R\sin\psi_e)(v_e \\
&\quad + u_R\sin\psi_e + v_R\cos\psi_e) + N_r(r_e + r_R) \\
&\quad + D_r(r_e + r_R)|r_e + r_R| + M_\psi\dot{r}_R + M_\psi\tau_r. \qquad (30)
\end{aligned}$$

By defining $\boldsymbol{\tau}_e = [\tau_u, \tau_v, \tau_r]^{\mathrm{T}}$ and substituting (28)–(30) into (27), together with (12) it yields the overall error dynamics for the AUV tracking control:

$$\dot{\mathbf{x}}_e = \begin{bmatrix} \dot{x}_e \\ \dot{y}_e \\ \dot{\psi}_e \\ \dot{u}_e \\ \dot{v}_e \\ \dot{r}_e \end{bmatrix} = \begin{bmatrix} y_e r_R - u_e + y_e r_e \\ -x_e r_R - v_e - x_e r_e \\ -r_e \\ \tau_u \\ \tau_v \\ \tau_r \end{bmatrix} \triangleq \mathbf{f}_e(\mathbf{x}_e, \boldsymbol{\tau}_e).$$

Appendix B
Proof of Theorem 1

Proof: Let $V(t, \mathbf{x}_e(t_0))$ denote the value function of the performance index. The state trajectory determined by $\bar{\boldsymbol{\tau}}_e(t)$ is denoted by $\bar{\mathbf{x}}_e(t)$; the state trajectory determined by the initial guess (17) is $\hat{\mathbf{x}}_e(t)$ and the corresponding cost is $\hat{V}(t, \mathbf{x}_e)$.

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1) For $0 \le t_1 \le t_2 < \delta$, V is nonincreasing since

$$V(t_2, \mathbf{x}_e(t_0)) = V(t_1, \mathbf{x}_e(t_0)) - \int_{t_1}^{t_2} \ell(\mathbf{x}_e(s), \boldsymbol{\tau}_e(s))ds \le V(t_1, \mathbf{x}_e(t_0)).$$

2) For $0 \le t_1 < \delta$, and $\delta \le t_2 < 2\delta$, we have

$$\begin{aligned} &V(t_2, \mathbf{x}_e(t_0+\delta)) - V(t_1, \mathbf{x}_e(t_0)) \\ &= V(t_2, \bar{\mathbf{x}}_e(t_0+\delta)) - V(t_1, \mathbf{x}_e(t_0)) \\ &\le \hat{V}(t_2, \bar{\mathbf{x}}_e(t_0+\delta)) - V(t_1, \mathbf{x}_e(t_0)) \\ &= h(\hat{\mathbf{x}}_e(\delta+T)) - h(\bar{\mathbf{x}}_e(T)) \\ &+ \int_{t_2}^{T+\delta} \ell(\hat{\mathbf{x}}_e(s), \hat{\boldsymbol{\tau}}_e(s))ds - \int_{t_1}^{T} \ell(\bar{\mathbf{x}}_e(s), \bar{\boldsymbol{\tau}}_e(s))ds. \end{aligned}$$

Observing (17), we shall notice that for $s \in [t_2, T]$, $\hat{\mathbf{x}}_e(s) = \bar{\mathbf{x}}_e(s)$, and $\hat{\boldsymbol{\tau}}_e(s) = \bar{\boldsymbol{\tau}}_e(s)$, thus we have

$$\begin{aligned} &V(t_2, \mathbf{x}_e(t_0+\delta)) - V(t_1, \mathbf{x}_e(t_0)) \\ &\le h(\hat{\mathbf{x}}_e(\delta+T)) - h(\hat{\mathbf{x}}_e(T)) \\ &+ \int_{T}^{T+\delta} \ell(\hat{\mathbf{x}}_e(s), \hat{\boldsymbol{\tau}}_e(s))ds - \int_{t_1}^{t_2} \ell(\bar{\mathbf{x}}_e(s), \bar{\boldsymbol{\tau}}_e(s))ds. \end{aligned}$$

Since we have $\dot{h}(\hat{\mathbf{x}}_e) + \ell(\hat{\mathbf{x}}_e, \hat{\boldsymbol{\tau}}_e) \le 0$, integrating both sides from T to $\delta + T$, then

$$h(\hat{\mathbf{x}}_e(\delta+T)) - h(\hat{\mathbf{x}}_e(T)) + \int_{T}^{T+\delta} \ell(\hat{\mathbf{x}}_e(s), \hat{\boldsymbol{\tau}}_e(s))ds \le 0.$$

Hence we have

$$\begin{aligned} &V(t_2, \mathbf{x}_e(t_0+\delta)) - V(t_1, \mathbf{x}_e(t_0)) \\ &\le -\int_{t_1}^{t_2} \ell(\bar{\mathbf{x}}_e(s), \bar{\boldsymbol{\tau}}_e(s))ds = -\int_{t_1}^{t_2} \ell(\mathbf{x}_e(s), \boldsymbol{\tau}_e(s))ds. \end{aligned}$$

3) For $k_1\delta \le t_1 < (k_1+1)\delta$ and $k_2\delta \le t_2 < (k_2+1)\delta$, with $k_1 < k_2$, by 2) the following inequality holds:

$$V(t_2) \le V(k_2\delta) \cdots \le V((k_1+1)\delta) \le V(t_1)$$

which implies

$$\begin{aligned} V(t, \mathbf{x}_e(t)) - V(t_0, \mathbf{x}_e(t_0)) &\le -\int_{t_0}^{t} \ell(\mathbf{x}_e(s), \boldsymbol{\tau}_e(s))ds \\ = -\int_{t_0}^{t} \mathbf{x}_e^{\mathrm{T}}\mathbf{Q}\mathbf{x}_e + \boldsymbol{\tau}_e^{\mathrm{T}}\mathbf{R}\boldsymbol{\tau}_e ds &\le -\int_{t_0}^{t} \mathbf{x}_e^{\mathrm{T}}\mathbf{Q}\mathbf{x}_e ds. \end{aligned}$$

By construction, we always have $V \ge 0$. Then we can say V is nonincreasing and lower bounded by zero. According to Barbalat's lemma [43], we have $\mathbf{x}_e^{\mathrm{T}}\mathbf{Q}\mathbf{x}_e \to 0$ as $t \to \infty$. Since $\mathbf{Q}$ is positive definite, the system (14) will be steered to the equilibrium point $(\mathbf{0}, \mathbf{0})$ asymptotically by the MPC controller, i.e., $\mathbf{x}_e \to 0$ as $t \to \infty$. ■

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